

Week 2: Homogeneous Transformations

EE0849 Introduction to Robotics - Detailed Notes

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1 Rotation Composition (Prerequisite)

1.1 Composition Rules

- **Current frame axis** -> post-multiply

$$R_{final} = R_{current}R_{new}$$

- **Fixed frame axis** -> pre-multiply

$$R_{final} = R_{new}R_{current}$$

1.2 Non-Commutativity

$$R_1 R_2 \neq R_2 R_1$$

This is why rotation order must be tracked carefully.

2 Homogeneous Transformations (Main Topic)

2.1 Rigid Motion as Affine Map

$$p^0 = R_1^0 p^1 + d_1^0$$

This combines rotation and translation but is not a single linear map in 3D coordinates.

2.2 Homogeneous Representation

$$H_1^0 = \begin{bmatrix} R_1^0 & d_1^0 \\ 0_{1 \times 3} & 1 \end{bmatrix} \in SE(3)$$

$$P = \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}, \quad P^0 = H_1^0 P^1$$

2.3 Composition Rule

$$H_2^0 = H_1^0 H_2^1$$

Interpret right-to-left in frame chains.

2.4 Inverse Transform

$$H^{-1} = \begin{bmatrix} R^T & -R^T d \\ 0 & 1 \end{bmatrix}$$

Important:

$$H^{-1} \neq H^T$$

2.5 Similarity Transformations

When a linear operator A is expressed in different frames, the representations are related by a similarity transform:

$$A^1 = (R_1^0)^{-1} A^0 R_1^0 = (R_1^0)^T A^0 R_1^0$$

Key property: **eigenvalues are preserved** under similarity transformations. This means physical quantities like natural frequencies or principal moments are frame-independent, even though the matrix representation changes.

2.6 $SE(3)$: Special Euclidean Group

The set of all rigid-body transformations in 3D forms the **Special Euclidean Group**:

$$SE(3) = \{(R, d) : R \in SO(3), d \in \mathbb{R}^3\}$$

- **6 DOF total:** 3 rotational (from R) + 3 translational (from d)
- **Not a vector space:** elements are composed by matrix multiplication, not addition
- **Foundation of robot kinematics:** every joint transform, tool frame, and sensor frame is an element of $SE(3)$

In matrix form, every element of $SE(3)$ is a 4×4 homogeneous transformation matrix:

$$H = \begin{bmatrix} R & d \\ 0_{1 \times 3} & 1 \end{bmatrix}$$

3 Orientation Parameterizations (Short Core)

3.1 Why Parameterize?

The orientation block R in H has 3 DOF but 9 entries under constraints, so compact parameter sets are useful.

3.2 Euler Angles (Z-Y-Z)

$$R = R_z(\phi)R_y(\theta)R_z(\psi)$$

Useful but can become singular.

3.3 Gimbal Lock

When $\sin \theta = 0$ (i.e., $\theta = 0^\circ$ or 180°), angle recovery is non-unique.

Only combinations (for example, $\phi + \psi$) remain identifiable.

3.4 Roll-Pitch-Yaw

A common robotics sequence is

$$R = R_z(\psi)R_y(\theta)R_x(\phi)$$

4 Python Connection

Use `spatialmath` S03 and SE3 objects to:

1. Build transforms
2. Compose frame chains
3. Compute inverses
4. Verify results numerically against hand calculations